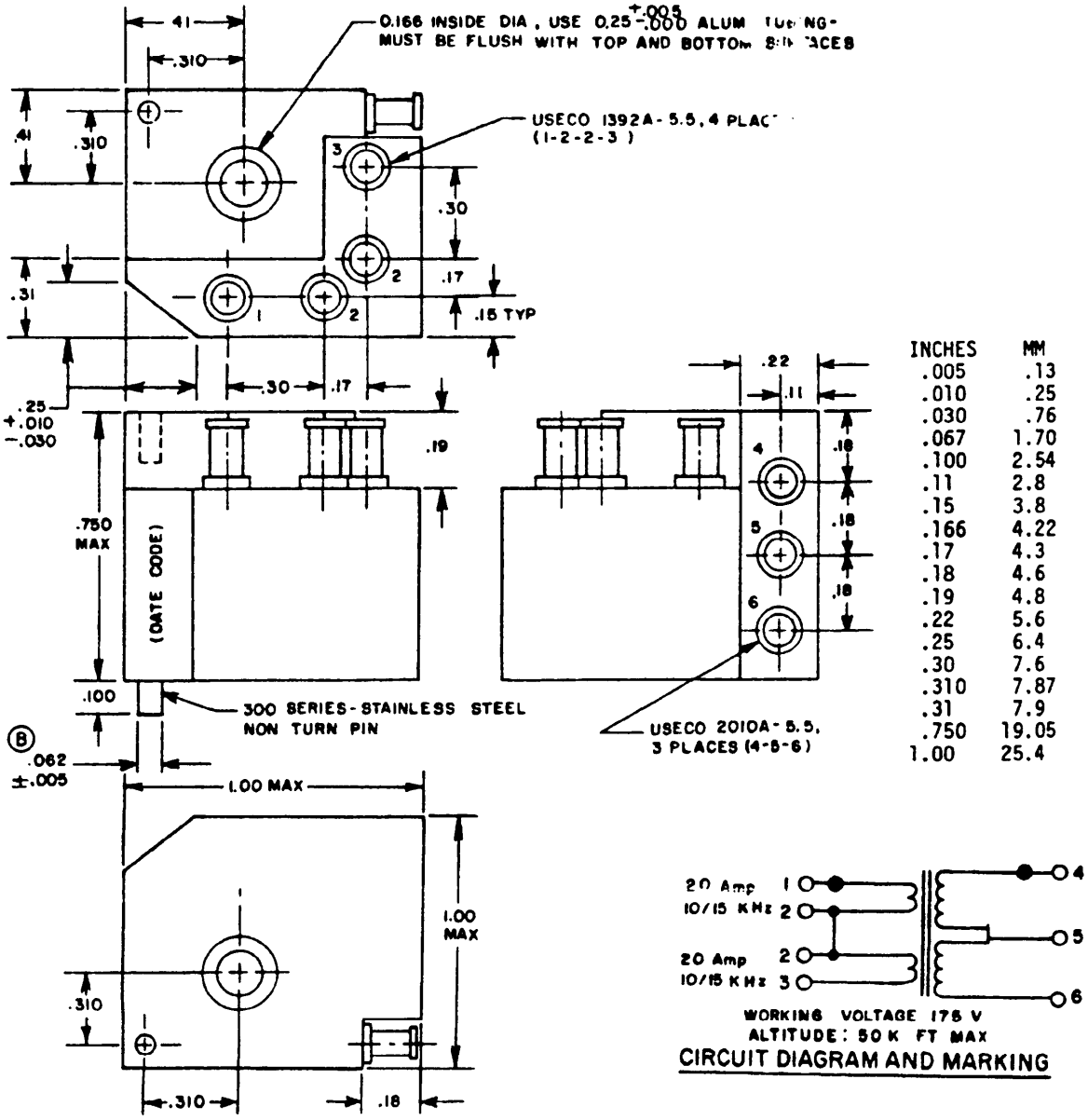
MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, 0.6 VA, 10/15 kHz, TF5V03ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the transformer described herein shall consist of this specification and the latest issue of MIL-T-27.



INCHES	MM
.005	.13
.010	.25
.030	.76
.067	1.70
.100	2.54
.11	2.8
.15	3.8
.166	4.22
.17	4.3
.18	4.6
.19	4.8
.22	5.6
.25	6.4
.30	7.6
.310	7.87
.31	7.9
.750	19.05
1.00	25.4

CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on side of case.

FIGURE 1. Dimensions and configurations.

(B) denotes changes

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FSC 5950

REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary current: 20 A at 10 to 15 kHz.

Power level: .60 W.

Secondary voltage: 6.1 V maximum.

Secondary current: .195 A.

Working voltage: 175 V.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Hysol C18 plastic (molded unit).

Terminals (solder lug):

(1-2-2-3): Capable of accepting two each AWG # 16 wires.

(4-5-6): Capable of accepting two each AWG # 20 wires.

Terminal height: .850 inch maximum.

Weight: 40 grams maximum.

Altitude: 50,000 feet maximum.

Operating temperature range: -55°C to +155°C.

Terminal strength: MIL-STD-202, method 211, test condition E, 5 pounds.

Dielectric withstanding voltage:

At sea level: 500 V rms.

At reduced barometric pressure: 300 V rms.

Electrical characteristics:

(B) No load: With 6.1 V rms, 7,000 Hz sine wave across (4-6), current in (4-6) shall be 0.0014 A maximum.

Voltage ratio V(1-3):

$\frac{V(1-3)}{V(4-6)}$ 0.0100 ±2 percent.

$\frac{V(2-3)}{V(1-3)}$ 0.5000 ±1 percent.

$\frac{V(5-6)}{V(4-6)}$ 0.5000 ±1 percent.

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Rated load:

Condtion 1: With 20.0 A ±0.5 percent, 10 kHz ±1 percent sine wave current applied to (1-2).

Current (4-5): 0.195 +0.005, -0.015 A with (4-5) loaded with 15 ohms ±5.0 percent and (5-6) not loaded.

Current (5-6): 0.195 ±0.005, -0.015 A with (5-6) loaded with 15 ohms ±5.0 percent and (4-5) not loaded.

Condition 2: With 20.0 A ±5 percent, 10 kHz ±1.0 percent sine wave current applied to (2-3).

Current (4-5): 0.195 +0.005, -0.015 A with (4-5) loaded with 15 ohms ±5.0 percent and (5-6) not loaded.

Current (5-6): 0.195 +0.005, -0.005 A with (5-6) loaded with 15 ohms ±5.0 percent and (4-5) not loaded.

DC resistance:

(1-3): 0.001 ohm maximum.

(4-6): 1.9 ohms maximum.

Center-tap balance (at low levels): ±1 percent.

Turns ratio: 1:100 ±2 percent.

Polarity: Additive with terminals (3-4) connected.

Saturation: See figure 2.

Temperature rise: 40°C with 10/15 kHz across (1-3) at an ambient temperature of 115°C.

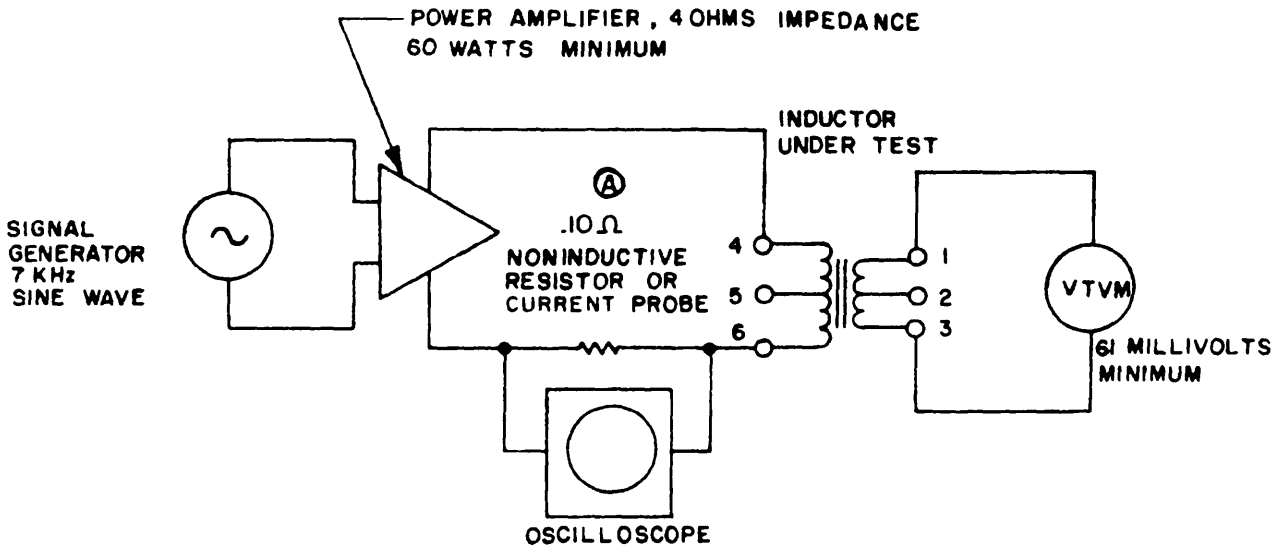
Vibration, high frequency: MIL-STD-202, method 204, test condition D.

Shock (specified pulse): MIL-STD-202, method 213, test condition A.

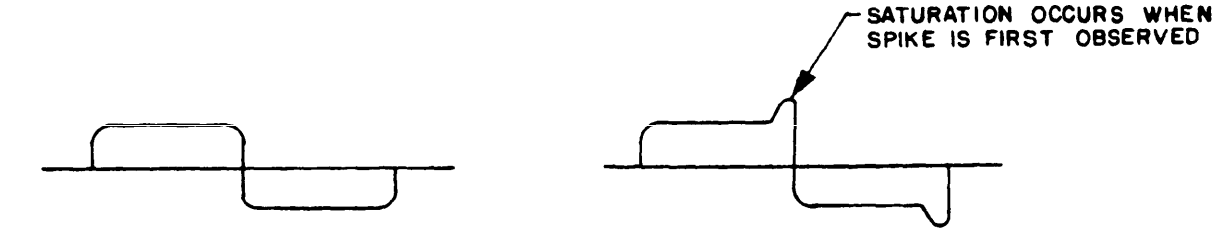
Marking location: See figure 1.

Part number: M27/270-01.

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1. Set signal generator to 7 KHz or less.
2. Adjust amplifier so that 61 millivolts minimum is across terminals (1 and 3).
3. The magnetizing current waveforms observed on the oscilloscope shall show no signs of saturation. See figure 2a and 2b.



2a. MAGNETIZING WAVEFORM
UNSATURATED

2b. MAGNETIZING WAVEFORM
SATURATED

FIGURE 2. Saturation test circuit.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5950-0641-6)

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